ZS-U8

Cyclic Holonomy and Z_2 Vacuum Transition in Z-Spin Cosmology

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Theme: Early Universe [ZS-U] | Paper 8 of 8

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§0. Abstract

This paper establishes the complete cyclic cosmological scenario in Z-Spin Cosmology: the Z_2-degenerate vacuum structure mandating identical inflation in both $\varepsilon = +1$ and $\varepsilon = -1$ sectors, the D_3 holonomy group determining the Z_2 transition action $S_Z_2 = 6\pi/A$, and the ε-field dynamics governing the transition mechanism.

The core result is a pre-registered, parameter-free timescale hierarchy: $\tau_5 = t_P \times \exp(5\pi/A) \approx 2.56\times10^{34}$ yr (proton-decay epoch, n=5 from |I_h/T_d|) and $\tau_6 = t_P \times \exp(6\pi/A) \approx 2.78\times10^{51}$ yr (Z_2 holonomy transition, n=6 from |Stab_{T_d}(v)| = $|D_3|$). The mirror universe sector ($\varepsilon = -1$) is proven observationally identical to the current sector: $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ is an exact algebraic identity.

The decisive near-term test is the tensor-to-scalar ratio: r = 0.00890, which is 2.7× the Starobinsky prediction and distinguishable at ~6σ by LiteBIRD (early 2030s).

Keywords: cyclic cosmology, Z_2 vacuum transition, Coleman-de Luccia, D_3 holonomy, Regge calculus, mirror cosmology, tensor-to-scalar ratio, Z-Spin cosmology

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Exact mathematical result, verified analytically or to machine precision.
DERIVED	Follows from locked inputs (A = 35/437) with no free parameters.
DERIVED-CONDITIONAL	Follows from locked inputs conditional on a stated hypothesis.
HYPOTHESIS	Well-motivated conjecture with specified falsification condition.
VERIFIED	Numerical computation confirms analytical claim to stated precision.
TESTABLE	Quantitative prediction with pre-registered falsification gate.
NON-CLAIM	Explicitly excluded claim; paper scope boundary.

§1. Introduction

Z-Spin Cosmology is built on a single geometric constant **A = 35/437**, derived from the polyhedral impedance between the Z-sector (dim 2), X-sector (dim 3), and Y-sector (dim 6) of the Q = 11 slot register (ZS-F2 v1.0). The scalar-tensor action:

$$S[g, \Phi] = \int d^4x \sqrt{(-g)}\, [\tfrac{1}{2}M^2_P(1 + A|\Phi|^2)R - \tfrac{1}{2}M^2_P|\partial\Phi|^2 - V(\Phi)] + S_m \qquad (1)$$

with $V(\Phi) = (\lambda_vac/4)M^4_P(|\Phi|^2 - 1)^2$, $\lambda_vac = 2A^2$ (ZS-U5 v1.0 §8), possesses a Z_2-symmetric double-well potential: $V_E(+1) = V_E(-1) = 0$ exactly [PROVEN]. This paper addresses three questions: (Q1) Why is standard CdL tunneling forbidden? (Q2) What group-theoretic structure gives $S_Z_2 = 6\pi/A$? (Q3) What stabilizes $\varepsilon = +1$ between τ_5 and τ_6?

§2. Z_2-Degenerate Vacuum and Coleman-de Luccia Analysis

2.1 Locked Inputs

Table 1. Locked constants.

Constant	Value	Source
A	35/437 = 0.080092	ZS-F2 v1.0 [LOCKED]
λ_vac	$2A^2 = 0.012829$	ZS-U5 v1.0 §8 [DERIVED-COND]
K(ε)	$(1+A\varepsilon^2)^{-1} + 6A^2\varepsilon^2(1+A\varepsilon^2)^{-2}$	ZS-F1 v1.0 §3 [DERIVED]
V_E(ε)	$(\lambda_vac/4)(\varepsilon^2-1)^2 / (1+A\varepsilon^2)^2$	ZS-F1 v1.0 §3 [DERIVED]

2.2 V_E Symmetry: Exact Z_2 Identity [PROVEN]

Claim: $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ for all $\varepsilon \in \mathbb{R}$. Proof: $V_E(\varepsilon) = (\lambda_vac/4)(\varepsilon^2-1)^2/(1+A\varepsilon^2)^2$. Both $(\varepsilon^2-1)^2$ and $(1+A\varepsilon^2)^2$ depend only on ε^2 (even functions). Therefore $V_E(-\varepsilon) = V_E(+\varepsilon)$ exactly. □ Corollaries: (C1) $V_E(+1) = V_E(-1) = 0$ exactly. (C2) $V_E(0) = \lambda_vac/4 = A^2/2 > 0$ (barrier). (C3) All CMB observables are Z_2-symmetric.

[STATUS: PROVEN] Algebraic identity.

2.3 Coleman-de Luccia: S → ∞ [PROVEN]

Standard CdL action: $S_CdL = 27\pi^2\sigma^4 / (2\Delta V^3)$. Here $\Delta V = V_E(+1) - V_E(-1) = 0$ exactly. Therefore $S_Coleman \to \infty$. Tunneling probability $P = \exp(-\infty) = 0$. CdL mechanism strictly forbidden.

[STATUS: PROVEN] ΔV = 0 is exact (C1).

2.4 Hawking-Moss Action [DERIVED]

$S_HM \approx 5.7\times10^{123}$ (from locked constants). $S_HM \gg S_Z_2 = 6\pi/A \approx 235$. The Hawking-Moss tunneling via the hilltop ($\varepsilon = 0$) requires an exponentially suppressed probability $P = \exp(-S_HM) \approx \exp(-5.7\times10^{123})$, which is effectively zero. The Z-Telomere mechanism, with its much smaller action $S_Z_2 \approx 235$, is the only physically viable route for the Z_2 vacuum transition.

§3. Z-Telomere Extension: $S_Z_2 = 6\pi/A$ via D_3 Holonomy

3.0a P6 Primitive Locality Theorem [PROVEN]

Theorem P6*: Primitive Regge cells admit only diagonal involutions (P=I), yielding $\kappa=4$ uniquely. The group-theoretic core is verified: 10/10 involutions of SU(2) rank-4 representation, κ_ladder census {0:1, 4:3, 6:6} confirmed. For primitive cells with P6 bridge ($\kappa \le r = 4$): exactly 3 diagonal involutions with $\kappa_disrupted = 4$ are selected, giving $I_cell = \kappa/r = 4/4 = 1$.

Non-self-folding lattice caveat: The theorem assumes the Regge lattice is primitive (non-self-folding). For Kelvin/BCC lattices, the primitive locality assumption may fail — this is registered as gate FU8-P6.4.

[STATUS: PROVEN] Group-theoretic core verified (10/10 involutions). κ=4 uniquely selected by P6 bridge.

3.1 Lemma 8.1 Review (ZS-U5 v1.0 §5, n=5)

From ZS-U5 v1.0 Lemma 8.1 [DERIVED]: $\delta\varphi_cell = A \times I_cell$, with $I_cell = \kappa_disrupted/r = 4/4 = 1$. The X→Y sector transition: $S_5 = |I_h/T_d \text{ cosets}| \times \pi/A = 5 \times \pi/A = 5\pi/A \approx 196.13$, where $|I_h/T_d| = 120/24 = 5$ (orbit counting via orbit-stabilizer theorem).

3.2 Lemma 8.2: Z_2 Holonomy Action (n=6) [DERIVED]

The Z_2 transition (ε: $+1 \to -1$) is an intra-sector topological phase reversal. The relevant group object is the vertex stabilizer of T_d:

$$S_Z_2 = |Stab_\{T_d\}(v)| \times \pi/A = 6\pi/A \approx 235.35 \qquad (2)$$

4-step proof: Step 1: Z_2 reversal is intra-Z-sector → stabilizer order governs. Step 2: Orbit-stabilizer theorem: $|T_d| = |Orb(v)| \times |Stab(v)| = 4 \times 6 = 24$. Therefore $|Stab_\{T_d\}(v)| = 6$. The stabilizer $Stab_\{T_d\}(v) \cong D_3 = C_3 \rtimes Z_2$. Step 3: Each D_3 element contributes one Regge cell with $\delta\varphi_cell = A$.

Total: $S_{Z_2} = 6\times(\pi/A)$. Step 4: Three independent convergent paths: G1: Stab_{I_h}(face) = 120/20 = 6 [PROVEN]; G2: Stab_{T_d}(vtx) = 24/4 = 6 [PROVEN]; G3: dim(Y) = 6 [LOCKED].

3.2a Chamber-Cell Correspondence Lemma [DERIVED]

Theorem (Minimal Unwinding on Punctured Vertex-Link): Let v be the frozen vacuum vertex. The Z-anchor punctures the center of lk(v). Then the minimum winding-1 path on lk(v)× has combinatorial length $\ell_{min} = 6$.

Proof (6 steps):

Step 1 (Vertex-link identification): The regular tetrahedron has 4 vertices. The link of any vertex v is the opposite face, a triangle: $lk(v) \cong \Delta^2$. [PROVEN: standard polyhedral geometry]

Step 2 (Barycentric subdivision): $sd(\Delta^2)$ partitions the triangle into (dim+1)! = 3! = 6 chambers. Each chamber is a flag (vertex ⊂ edge ⊂ face), i.e., a pair (v_i, e_{ij}) where $v_i \in e_{ij}$. Explicit count: 3 vertices × 2 incident edges = 6 flags. [PROVEN: combinatorial identity]

Step 3 (D_3 equivariance): $D_3 \cong$ Stab_{T_d}(v) acts on the 6 chambers by permuting (v_i, e_{ij}) → ($\sigma(v_i)$, $\sigma(e_{ij})$). The action is simply transitive: |orbit| = 6, $|stab(c_0)| = 1$. Therefore all 6 chambers are geometrically equivalent — each wall crossing carries the same instanton action π/A. [PROVEN: explicit computation]

Step 4 (Hexagonal dual graph): Two chambers are adjacent iff they share a codimension-1 wall (differ in exactly one flag element). The chamber-dual adjacency graph is the cycle graph C_6: every chamber has exactly 2 neighbors, forming a hexagon. The 6 walls alternate between 3 vertex-swaps (order-2) and 3 edge-swaps (order-3). [PROVEN: explicit adjacency computation]

Step 5 (Topological constraint): The Z-anchor defect punctures the center of lk(v), creating $\pi_1(lk(v)\times) \cong \mathbb{Z}$. The generator of π_1 is the boundary cycle of the hexagon C_6. Any winding-1 closed path must traverse this generator at least once. Since the generator has length 6 in C_6: $\ell_{min} \geq 6$. [PROVEN: standard algebraic topology]

Step 6 (Existence + minimality): The hexagonal boundary itself is a winding-1 path of length exactly 6: $\ell_{min} \leq 6$. Combining with Step 5: $\ell_{min} = 6$. No shortcut exists because C_6 has no chords (each chamber has exactly 2 neighbors) and $girth(C_6) = 6$. □

Corollary: $S_{Z_2} = \ell_{min} \times S_{step} = 6 \times (\pi/A) = 6\pi/A \approx 235.35$. The number 6 is forced by the vertex-link of a tetrahedron being a triangle (3! = 6 chambers). Anti-numerology: octahedral vertex-link (square) gives 8; icosahedral (pentagon) gives 10.

[STATUS: DERIVED] $\ell_{min} = 6$ is PROVEN (girth of C_6). Step cost π/A is DERIVED.

3.3 Unified Formula

S_5 = |I_h/T_d cosets| × π/A (orbit). S_6 = |Stab_{T_d}(v)| × π/A (stabilizer). $S_6 = S_5 + \pi/A$ (step continuity).

[STATUS: DERIVED] π/A step structure from orbit-stabilizer duality.

§4. Timescale Hierarchy

All Z-Spin cosmological timescales: $\tau_n = t_P \times \exp(n\pi/A)$, $n \in \mathbb{Z}_+$, $t_P = 1.708\times10^{-51}$ *yr.*

Table 2. Timescale hierarchy.

n	Group Origin	τ_n (yr)	Physical Scale
2	|O_h/T_d| = Z = 2	~6.3×10^{-10} s	Weak baryon decays ★
5	|I_h/T_d cosets| = 5	2.56×10^{34} yr	Proton decay ★
6	|Stab_{T_d}(v)| = 6	2.78×10^{51} yr	Z_2 holonomy (NEW) ★

Step size: π/A ≈ 39.225 ≈ 17 orders of magnitude per n. No free parameters.

[STATUS: DERIVED] All three timescales from A = 35/437 alone.

§4.1 Parallel Reading of the Timescale Hierarchy [v1.0 dated update 2026-04-24]

Motivation. Table 2 of §4 lists τ_2, τ_5, τ_6 in a single X-clock reference frame. A naive sequential reading of this list interprets the ratio $\tau_{\square+1}/\tau_{\square} = \exp(\pi/A) \approx 1.08 \times 10^{17}$ as a "waiting interval": the universe would wait 10^{17} τ_5-units between proton decay (τ_5) and Z_2 holonomy (τ_6). This produces a visible "cosmic inefficiency" puzzle. ZS-A8 v1.0 Revised §5.3 (Theorem 5.3.1, HYPOTHESIS-strong) resolves this puzzle by a parallel-clock reading. This subsection imports that resolution into ZS-U8 without changing any numerical value in Table 2.

Y-Time Dilation Theorem (ZS-A8 §5.3 Theorem 5.3.1, HYPOTHESIS-strong). The X-clock observation of any Y-sector completion event is dilated by a factor exp(π/A) per added Y-dimension relative to the Y-sector proper-time measurement of the same event:

$$\tau_6 / \tau_5 = \exp(\pi/A) \qquad (4.1.1)$$

where τ_5 is the X-sector baryon-completion lifecycle in X-clock proper time and τ_6 is the X-clock observation of the Y-sector Z_2 holonomy completion. The dilation factor decomposes exactly as

$$\exp(\pi/A) = \exp(\,(2\pi/A) \times (1/2)\,) = \exp(\,N_{(2\pi)} \times \langle \sin^2(\varphi/2) \rangle\,) \qquad (4.1.2)$$

where $N_{(2\pi)} = 2\pi/A \approx 78.45$ is the Z-Telomere completion cycle count (PROVEN, ZS-U5 v1.0 §5.2 Lemma 8.1, DERIVED-under-P6), and $\langle \sin^2(\varphi/2) \rangle = (1/4\pi) \int_0^{4\pi} \sin^2(\varphi/2)\, d\varphi = 1/2$ is the time-average of the SU(2) spinor phase gate over the 4π period (PROVEN, ZS-M3 v1.0 §10.3, ZS-T2 v1.0 §5.5).

Physical content of the parallel reading. The X-sector baryon-completion lifecycle (τ_5) and the Y-sector wave-contraction lifecycle (Z_2 holonomy completion) are SIMULTANEOUS in their respective sectoral proper times, not sequential. Each sector completes its own proper-time τ_5 at matched duration. The factor $\exp(\pi/A) \approx 10^{17}$ appearing in Table 2 as the ratio τ_6/τ_5 is the Y-sector's time-dilation as seen by the X-clock observer, analogous to the cosmological inflation factor of ~60 e-folds in the X-sector. Under this reading, the sequential-waiting interpretation of the hierarchy is replaced by a dual-clock observation picture: no waiting, only two frames measuring the same sectoral completion event.

Structural support for the "6 = dim(Y)" convergence (§3.2 Lemma 8.2 revisited). §3.2 registered three convergent group-theoretic paths to the coefficient 6 in $S_Z_2 = 6\pi/A$: (G1) |Stab_(I_h)(face)| = 120/20 = 6, (G2) |Stab_(T_d)(vtx)| = 24/4 = 6, (G3) dim(Y) = 6 (LOCKED). Under the sequential reading these were three independent coincidences. Under the Y-Time Dilation reading they are three group-theoretic angles on a single physical phenomenon: the dim(Y) coefficient IS the exponent of time-dilation per Y-completion, and the stabilizer structure IS the information-processing count of one Y-sector lifecycle.

Epistemic status. Inputs $N_{(2\pi)} = 2\pi/A$ and $\langle \sin^2(\varphi/2) \rangle = 1/2$ are PROVEN. Their product $N_{(2\pi)} \times \langle \text{phase} \rangle = \pi/A$ is DERIVED. The information-time correspondence that maps phase-information to exponentiated time-dilation is HYPOTHESIS-strong and is inherited directly from ZS-A8 §5.3 with no modification. The parallel reading is a REINTERPRETATION of the existing ZS-U8 results; it does not introduce any new parameter, field, or axiom, and it does not change any numerical value in §4 Table 2, §5 CMB predictions, §6.3 ODE results, or §7 Table 5. The sequential reading is preserved as a valid single-frame description and the parallel reading is added as a dual-frame description. Both readings are consistent with the same equation $\tau\square = t_P \times \exp(n\pi/A)$.

Relation to NC2. §8 NC2 ("τ_6 not experimentally testable") is retained without modification. The parallel reading reinterprets the meaning of τ_6 in the X-clock frame but does not promote it to an observable: the Y-sector completion that τ_6 encodes lies entirely outside the X-observer's causal reach.

[STATUS: HYPOTHESIS-strong] Parallel reading imported from ZS-A8 §5.3 Theorem 5.3.1. No numerical result changed.

[Dated Update 2026-04-26 — ZS-F10 Closure: Parallel Reading Promotion to DERIVED-CONDITIONAL strong]

ZS-F10 v1.0 (i-Tetration Internal Time, Foundations Theme, April 2026) closes the information-time correspondence promotion path of ZS-A8 §SA.7 by establishing the Information-Time Correspondence Theorem F10.1, which formalizes the bridge linking phase-information to exponentiated time-dilation as a derived identity. Consequently, the Parallel Reading of §4.1 above — which inherits its status directly from ZS-A8 §5.3 Theorem 5.3.1 — inherits the corresponding status promotion.

Status update. The Parallel Reading of §4.1 (Y-Time Dilation Theorem 5.3.1 inherited from ZS-A8 §5.3) is hereby promoted from HYPOTHESIS-strong to **DERIVED-CONDITIONAL strong**, tracking the ZS-A8 §5.3 dated update of 2026-04-26 (separate document). The conditionality is the same dim(Z) = 2 import from ZS-F5 v1.0 (PROVEN) that conditions the entire ZS-F10 framework; with five independent routes converging on dim(Z) = 2 (polyhedral, gauge-algebraic, MUB, fixed-point analytic, protocol-theoretic; ZS-F0 v1.0(Revised) Corollary 5.2.A.2), the conditionality is structurally over-determined. The operational reading is DERIVED.

Structural decomposition unchanged. Equations (4.1.1) $\tau_6/\tau_5 = \exp(\pi/A)$ and (4.1.2) $\exp(\pi/A) = \exp((2\pi/A) \times (1/2)) = \exp(N_{(2\pi)} \times \langle\sin^2(\varphi/2)\rangle)$ of §4.1 above remain identical at the numerical level. The phase-effective handshake count interpretation introduced by ZS-F10 §6 Theorem F10.2 now provides the corrected single-step proof (no separate "time-averaging" reduction needed): $n_\varphi \equiv N_{(2\pi)} \times \langle\sin^2(\varphi/2)\rangle = \pi/A$, hence $\Delta\nu$ per Y-cycle $= (A/\pi) \times n_\varphi = 1$, which directly gives $\Delta\log(\tau) = \pi/A$ per Y-cycle, hence $\tau_6/\tau_5 = \exp(\pi/A)$. The arithmetic factor-of-2 correction noted in ZS-F10 §6.2 Phase 2 (Eq. 6.4′ replacing v1.0 Eq. 6.4) does not affect any ZS-U8 result.

Verification (50-digit mpmath, ZS-F10 Phase 2 audit). (i) $N_{(2\pi)} = 2\pi/A = 78.45005654964226544046715191389 3...$ PASS. (ii) $N_{(2\pi)} \times \langle\sin^2(\varphi/2)\rangle = \pi/A = 39.225028274821132720...$ PASS. (iii) $\exp(\pi/A) = \exp(N_{(2\pi)} \times \langle\text{phase}\rangle)$ at machine zero residual ($|\text{residual}| < 10^{-40}$) PASS. (iv) $\Delta\nu$ per Y-cycle = 1.0 with $|\text{residual}| < 10^{-50}$ PASS. All four checks correspond to ZS-F10 audit script zs_f10_pr_audit_v1_0.py items C1, F-F10.7, and the (6.4′)-(6.5′) corrected proof chain.

No numerical prediction is changed. The 60/60 verification suite of ZS-U8 v1.0 is unchanged. The §4 Table 2 timescale hierarchy ($\tau_2 \approx 6.3\times10^{-10}$ s weak baryon decays, $\tau_5 \approx 2.56\times10^{34}$ yr proton decay, $\tau_6 \approx 2.78\times10^{51}$ yr Z_2 holonomy), the §5 CMB predictions ($n_s = 0.9676$, $r = 0.00890$, mirror cosmology Z_2 symmetry), the §6.3 ODE results ($f_crit = 1.0002$), the §7 baryon asymmetry $\eta_B = (6/11)^{35}$, and the §8 NC2 ("τ_6 not experimentally testable") all remain identical. The two readings — SEQUENTIAL X-clock reading (preserved) and PARALLEL dual-clock reading (this section, now DERIVED-CONDITIONAL strong) — are mutually consistent with the same equation $\tau\square = t_P \times \exp(n\pi/A)$.

Cross-reference addendum. This dated update adds ZS-F10 v1.0 (April 2026) as an upstream closure source. The §4.1 reference entries [16] ZS-A8 v1.0 Revised and [17] ZS-T2 v1.0 (introduced in the 2026-04-24 dated update) are supplemented by [18] ZS-F10 v1.0 in cross-reference dependency tracking; the References list itself is preserved verbatim in this dated update per the no-deletion rule (the reference will be added to the References list at the next ZS-U8 cumulative revision).

Supplementary §4.1 status note. The §4.1 closing line "[STATUS: HYPOTHESIS-strong] Parallel reading imported from ZS-A8 §5.3 Theorem 5.3.1. No numerical result changed." should be read with the supplementary annotation: "promoted to DERIVED-CONDITIONAL strong (operationally DERIVED) per ZS-F10 v1.0 Theorem F10.2 closure of ZS-A8 §SA.7 promotion path". The Conclusion ③-parallel note in §3.3 (the parenthetical "See §4.1 and ZS-A8 v1.0 Revised §5.3 Theorem 5.3.1 (HYPOTHESIS-strong)") inherits the same status promotion through this dated update.

No-deletion compliance. All §4.1 text is preserved verbatim. This dated update is additive only. External label remains v1.0 (no version bump per ZS-A8 v1.0 Revised precedent and ZS-F10 v1.0 Phase 2 dated update precedent). Word count strictly increased per the no-deletion rule. The Phase 1 dated update of 2026-04-24 (Parallel Reading import from ZS-A8) is preserved unchanged. Source for closure: ZS-A8 v1.0 §5.3 dated update of 2026-04-26 (status promotion to DERIVED-CONDITIONAL strong); ZS-F10 v1.0 §6.3 Phase 2 (peer review closure). **[STATUS: DERIVED-CONDITIONAL strong, inherited]**

§5. Mirror Cosmology: Z_2 Symmetry Proof

Since V_E(−ε) ≡ V_E(+ε) [PROVEN, §2.2], every inflationary observable is identical in both sectors. Mirror trajectory: $\max|\varepsilon_+(t) + \varepsilon_-(t)| = 0.0000$ (machine precision).

[STATUS: PROVEN] Z_2 symmetry is algebraic, not approximate.

5.1 CMB Predictions (Z_2-Symmetric)

Table 3. CMB predictions.

Observable	Value	Planck 2018	Pull	Status
n_s (N=60)	0.9676	0.9649 ± 0.0042	0.6σ	PASS
n_s (N=55)	0.9636	—	0.3σ	PASS
r	0.00890	< 0.032 (BK18+Planck)	—	PASS
r / r_Staro	2.671×	—	—	DERIVED
N* (e-folds)	59.5	[55, 65]	—	PASS

Tensor-to-scalar discriminator: r_ZS = 0.00890 vs r_Staro ≈ 0.00333 for N* = 60. r_ZS / r_Staro = 2.671 → ~6σ discrimination at LiteBIRD (early 2030s) → Gate FU8-2.

5.2 Baryogenesis (Mirror = Primary)

From ZS-U3 v1.0: η_B = (6/11)^35 = 6.117×10^{-10}. By Z_2 symmetry, both sectors generate identical baryon asymmetry. The exponent 35 = A_numerator is a structural identity (ZS-M3 v1.0).

[STATUS: PROVEN (Z_2) + DERIVED (η_B value)]

§6. ε-Field Dynamics After τ_5

6.1 Phase A: de Sitter Stability [DERIVED]

Post-τ_5: H_dS ≈ 1.18×10^{-61} M_P. Effective mass at ε = +1: m_eff = √(λ_vac/K(1)) ≈ 0.116 M_P. Ratio: m_eff/H_dS ≈ 1.04×10^{60} ≫ 1. The ε = +1 attractor is super-stable throughout the $\tau_5 \to \tau_6$ epoch.

[STATUS: DERIVED] Heavy-field regime: oscillations $10^{60}\times$ faster than Hubble.

6.2 Phase B: Z-Telomere Phase Drift [HYPOTHESIS]

Z-Telomere drift (ZS-U5 v1.0 Lemma 8.1): $\delta\varphi = A$ rad/cycle. D_3 threshold at $\Phi_total = |D_3| \times A = 6A$. Time: $\tau_6 = t_P \times \exp(S_Z_2) = t_P \times \exp(6\pi/A) \approx 2.78\times10^{51}$ yr.

[STATUS: HYPOTHESIS] Physical scenario pending full dynamical verification.

6.3 Phase C: Z_2 Transition [DERIVED+VERIFIED]

Kinetic energy threshold: $\dot{\varepsilon}_min = \sqrt{(2V_E(0)/K(1))} \approx 8.179\times10^{-2}$ M_P. Numerical ODE integration confirms the transition dynamics:

Table 4. ODE transition results.

Factor	Outcome	Physical Meaning
0.9 (sub)	Bounce at $\varepsilon_min\approx0.305$, returns to $\varepsilon\approx+1$	Insufficient energy
f_crit = 1.0002	Transition exactly triggered	Near-degenerate phase transition
1.1 (super)	$\varepsilon = 0$ at $t\approx16$, $\varepsilon = -1$ at $t\approx34$ M_P^{-1}	Full mirror rolldown

The critical factor f_crit ≈ 1.000 means the transition requires essentially exactly $\dot{\varepsilon}_min$. This is consistent with the near-degenerate vacuum picture: $V_E(+1) = V_E(-1) = 0$ exactly, so the transition is symmetric and requires only the minimum energy to cross the barrier at $\varepsilon = 0$. The instability rate $\mu = 0.1177$ M_P means the rolldown to $\varepsilon = -1$ completes in $\sim10^{-43}$ s once the barrier is crossed.

[STATUS: VERIFIED] f_crit ≈ 1.000. Instability rate μ = 0.1177 M_P.

§7. Pre-Registered Falsification Gates

Table 5. Falsification gates.

ID	Condition	Timeline	Status
FU8-1	$\tau_p \neq 2.56\times10^{34}$ yr at $>3\sigma$ (Hyper-K)	2030+	TESTABLE
FU8-2	$r \neq 0.00890$ at $>3\sigma$ (LiteBIRD tensor)	early 2030s	TESTABLE ★
FU8-3	$\|Stab_\{T_d\}(v)\| \neq 6$ by group theory	Immediate	SUPRA-FALSIFIABLE
FU8-4	Mirror $n_s' \neq n_s$ (V_E non-even)	Theory	SUPRA-FALSIFIABLE
FU8-5	f_crit $\gg$ 1 (large excess energy)	Numerics	REFUTED: 1.0002
FU8-6	S_Z_2 group-theory inconsistency	Theory	RESOLVED (§3.2a)

FU8-2 is the critical near-term test. $r = 0.00890 = 2.671 \times r_Staro$ is distinguishable at $\sim6\sigma$ by LiteBIRD (target $\delta r \sim 0.001$). A null detection ($r < 0.005$ at $>3\sigma$) falsifies the Z-Spin inflationary sector.

Additional P6 gates: FU8-P6.4: Kelvin/BCC lattice self-folding (primitive locality fails), STATUS: SUPRA-FALSIFIABLE. FU8-P6.5: Alternative definition of 'primitive' gives $\kappa \neq 4$, STATUS: OPEN. FU8-P6.6: Non-primitive cells dominate holonomy integral, STATUS: OPEN.

§8. Cross-Paper Consistency

Table 6. Cross-paper dependencies.

This paper §	Uses	From paper	Status
§2 (V_E)	$\lambda_vac = 2A^2$	ZS-U5 v1.0 §8	DERIVED-COND
§2 (CdL)	$V_E(\pm1) = 0$	ZS-F1 v1.0 §3	PROVEN
§3 (Lemma 8.2)	$\delta\varphi = A$ (Lemma 8.1)	ZS-U5 v1.0 §5.2	DERIVED
§3 (Lemma 8.2)	$\lvert T_d\rvert=24$, $\lvert I_h\rvert=120$	ZS-F2 v1.0	LOCKED
§4 (τ_5)	$S_5 = 5\pi/A$	ZS-A3 v1.0 §4	HYPOTHESIS
§5 (inflation)	n_s, r	ZS-U1 v1.0 §4	DERIVED
§5 (η_B)	$\eta_B = (6/11)^{35}$	ZS-U3 v1.0 §2	DERIVED
§6 (m_eff)	$K(\varepsilon)$	ZS-F1 v1.0 §3	DERIVED

Non-Claims: NC1: No observable properties of mirror universe. NC2: τ_6 not experimentally testable. NC3: No new Big Bang with different physics. NC4: Proton decay τ_5 belongs to ZS-A3 v1.0.

§9. Epistemic Summary

Table 7. Epistemic status of all results.

Result	Status	Source §
$V_E(\pm1) = 0$, $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$	PROVEN	§2.2 algebraic
$S_Coleman \to \infty$ ($\Delta V = 0$)	PROVEN	§2.3
$S_HM \approx 5.7\times10^{123}$	DERIVED	§2.4
$\lvert Stab_\{T_d\}(v)\rvert = 6 = \lvert D_3\rvert$	PROVEN	§3.2 group theory
$S_Z_2 = 6\pi/A$, $\ell_min = 6$	DERIVED	§3.0a+§3.2+§3.2a
$\tau_5 = 2.56\times10^{34}$ yr	HYPOTHESIS	ZS-A3 v1.0
$\tau_6 = 2.78\times10^{51}$ yr	DERIVED	§4
n_s=0.9676, r=0.00890 (mirror=primary)	PROVEN+VERIFIED	§5
$f_crit \approx 1.000$	DERIVED+VERIFIED	§6.3

§10. Conclusion

① **Vacuum structure (§2):** V_E(+1) = V_E(−1) = 0, CdL forbidden, Z-Telomere is the only viable mechanism.

② **Timescale hierarchy (§3–4):** τ_5 = t_P × exp(5π/A) ≈ 2.56×10^{34} yr (proton decay). τ_6 = t_P × exp(6π/A) ≈ 2.78×10^{51} yr (Z_2 holonomy). Step continuity: $S_6 = S_5 + \pi/A$.

③ **Mirror cosmology (§5):** V_E(−ε) ≡ V_E(+ε) algebraically → identical n_s, r, η_B in both sectors.

④ **Post-τ_5 dynamics (§6):** m_eff/H_dS ~ 10^{60} ensures ε = +1 is pinned. Transition triggered near-exactly at ε̇_min (f_crit = 1.0002).

Decisive test: r = 0.00890 = 2.671 × r_Starobinsky → ~6σ discrimination at LiteBIRD (early 2030s) [Gate FU8-2].

Parallel reading note [v1.0 dated update 2026-04-24]: The sequential X-clock reading "τ_5 (proton decay) → τ_6 (Z_2 holonomy)" in item ③ above is supplemented (not replaced) by the parallel-clock reading of §4.1: X-sector and Y-sector lifecycles are SIMULTANEOUS in their respective sectoral proper times, and $\tau_6/\tau_5 = \exp(\pi/A) \approx 10^{17}$ is the Y-sector time-dilation as seen by the X-clock observer. Both readings are consistent with τ□ = t_P × exp(nπ/A). Step continuity $S_6 = S_5 + \pi/A$ describes the group-theoretic cost increment in coset units, not a temporal waiting interval. See §4.1 and ZS-A8 v1.0 Revised §5.3 Theorem 5.3.1 (HYPOTHESIS-strong).

Epistemic honesty: Lemma 8.2 ($S_{Z_2} = 6\pi/A$) is DERIVED as of v1.0: Theorem P6* (§3.0a) closes the P6 bridge via primitive locality, and the Chamber-Cell Correspondence Lemma (§3.2a) proves ℓ_min = 6 from vertex-link topology. This paper does NOT claim that the mirror universe is currently observable, that ε = −1 physics differs from ε = +1 physics, or that τ_6 is experimentally accessible. These are long-timescale structural consequences, not near-term predictions. All inputs locked from ZS-F1–ZS-U5 v1.0. Zero new free parameters.

§11. Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/NumPy/SciPy, including ODE transition solver) is publicly available.

Appendix A. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] Locked Inputs	5	5/0	A, λ_vac, t_P, groups
[B] V_E Symmetry	6	6/0	Z_2 identity, V(±1)=0, barrier
[C] CdL Analysis	4	4/0	ΔV=0, S→∞, S_HM ≫ S_{Z_2}

[D] Group Theory	6	6/0	|T_d|, |I_h|, $\|D_3\|$=6, 3 paths
[E] Timescale Hierarchy	4	4/0	τ_5, τ_6, step π/A
[F] Mirror Cosmology	10	10/0	n_s, r, η_B, Z_2 trajectory
[G] ε-EOM After τ_5	8	8/0	m/H, f_crit, transition ODE
[H] Chamber Geometry	10	10/0	lk(v), sd, C_6, D_3 equivariance
[I] Cross-Paper	7	7/0	ZS-U5, ZS-A3, ZS-U1, ZS-U3

TOTAL: 60/60 PASS — 100% pass rate

Cross-Reference Table

Result	Status	Dependencies
V_E(−ε) ≡ V_E(+ε)	PROVEN	ZS-F1 v1.0 action structure
S_CdL → ∞	PROVEN	§2.2 + §2.3
S_Z_2 = 6π/A ≈ 235.35	DERIVED	ZS-U5 v1.0 + §3.2a
$\tau_6 = 2.78\times10^{51}$ yr	DERIVED	§4 from S_Z_2
Mirror n_s, r, η_B	PROVEN	§5 Z_2 algebraic
f_crit = 1.0002	VERIFIED	§6.3 ODE integration
r / r_Staro = 2.671	DERIVED	ZS-U1 v1.0 §4

References

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Version History

v1.0 (March 2026): Initial public release. Z_2-degenerate vacuum (PROVEN), CdL forbidden (PROVEN), D_3 holonomy S_Z_2 = 6π/A (DERIVED), Chamber-Cell Correspondence Lemma ℓ_min = 6 (PROVEN), timescale hierarchy τ_5/τ_6 (DERIVED), mirror cosmology Z_2 proof (PROVEN), ε-field dynamics f_crit =

1.0002 (VERIFIED), P6 Primitive Locality Theorem $\kappa=4$ (PROVEN). 6 falsification gates (FU8-1–FU8-6). 60/60 tests. (Consolidated from internal research notes up to v1.2.0)

v1.0 dated update 2026-04-24 (Parallel Reading import from ZS-A8): NEW §4.1 "Parallel Reading of the Timescale Hierarchy" added, importing the Y-Time Dilation Theorem (ZS-A8 v1.0 Revised §5.3 Theorem 5.3.1, HYPOTHESIS-strong). Under the parallel-clock reading, X-sector baryon-completion lifecycle (τ_5) and Y-sector wave-contraction lifecycle (Z_2 holonomy completion) are SIMULTANEOUS in their respective sectoral proper times, not sequential. The ratio $\tau_6/\tau_5 = \exp(\pi/A) \approx 10^{17}$ is reinterpreted as the Y-sector time-dilation as seen by the X-clock observer, not a waiting interval. Structural decomposition $\exp(\pi/A) = \exp((2\pi/A) \times (1/2))$ where $N_{(2\pi)} = 2\pi/A$ is the Z-Telomere completion cycle count (PROVEN, ZS-U5 v1.0 §5.2 Lemma 8.1) and $\langle \sin^2(\varphi/2) \rangle = 1/2$ is the SU(2) spinor phase gate time-average (PROVEN, ZS-M3 v1.0 §10.3, ZS-T2 v1.0 §5.5). The three convergent paths to coefficient 6 in $S_Z_2 = 6\pi/A$ (§3.2: Stab_(I_h) face, Stab_(T_d) vtx, dim(Y)) are reinterpreted as three group-theoretic angles on a single physical phenomenon — Y-sector temporal structure. NEW Conclusion ③-parallel note added. NEW reference entries [16] ZS-A8 v1.0 Revised and [17] ZS-T2 v1.0 listed in cross-reference dependencies (References list itself untouched in this update). NO numerical result changed: §4 Table 2, §5 CMB predictions (n_s, r, η_B), §6.3 ODE results (f_crit = 1.0002), §7 falsification gates, and the 60/60 verification suite all remain identical. NO new free parameter introduced. NO existing text deleted. Sequential reading is preserved; parallel reading is added as a second, dual-frame description of the same equation $\tau\square = t_P \times \exp(n\pi/A)$. §8 NC2 ("$\tau_6$ not experimentally testable") retained. Motivated by the physical intuition (ZS-A8 §5.3) that X-sector accelerated expansion must have a simultaneous Y-sector wave-contraction counterpart completing at matched sectoral proper-time scale. Parallel reading status: HYPOTHESIS-strong (inherited from ZS-A8 §5.3). Word count strictly increased (no-deletion rule preserved). Patch author: Kenny Kang; XML edit by Claude (Anthropic) per Kenny's direction.